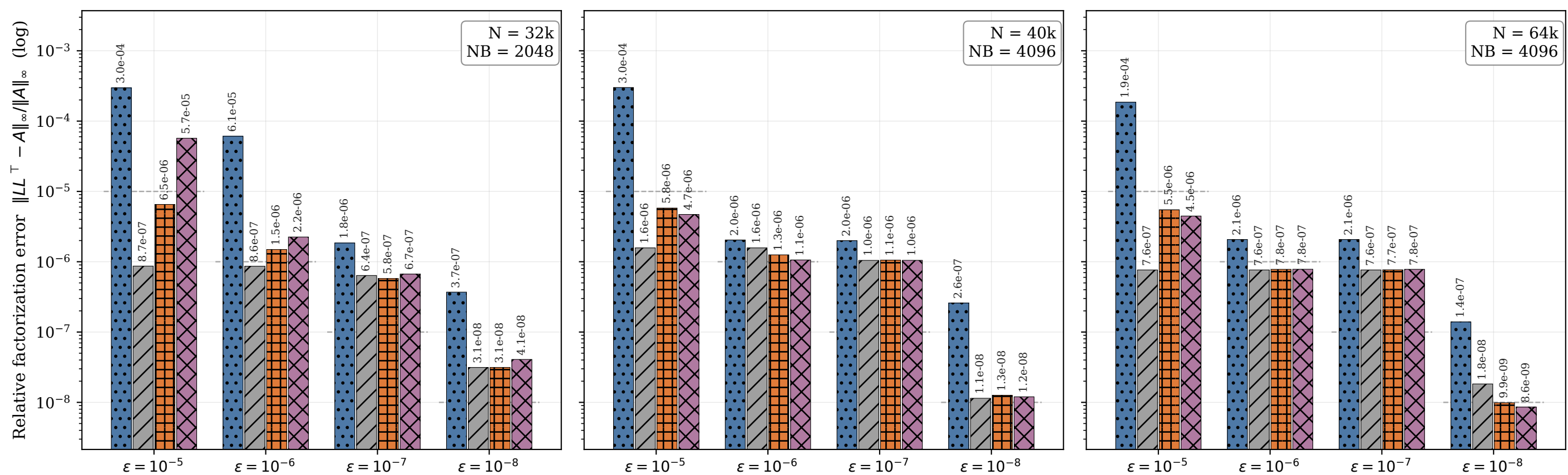




Bars within each cluster (left -> right) . grey dashed = source ϵ

Plain FP8 baseline (no MX; subnormal-allowed)

Full ladder tile (MXFP4 -> MXFP8 -> MXFP16 -> FP32 -> FP64)

IEEE ladder (FP8 E4M3 -> FP16 -> FP32 -> FP64; no MX)

Full ladder vec1D-32 (MXFP4 -> MXFP8 -> MXFP16 -> FP32 -> FP64)